



The Coin Change Problem

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Given an amount and the denominations of coins available, determine how many ways change can be made for amount. There is a limitless supply of each coin type.

Example

There are ways to make change for : , , and .

Function Description

Complete the `getWays` function in the editor below.

`getWays` has the following parameter(s):

- `int n`: the amount to make change for
- `int c[m]`: the available coin denominations

Returns

- `int`: the number of ways to make change

Input Format

The first line contains two space-separated integers and , where:

 is the amount to change

 is the number of coin types

The second line contains space-separated integers that describe the values of each coin type.

Constraints

-
-
-
- Each is guaranteed to be distinct.

Hints

Solve overlapping subproblems using [Dynamic Programming \(DP\)](#):

You can solve this problem recursively but will not pass all the test cases without optimizing to eliminate the [overlapping subproblems](#). Think of a way to store and reference previously computed solutions to avoid solving the same subproblem multiple times. * Consider the degenerate cases:

- How many ways can you make change for cents? - How many ways can you make change for cents if you have no coins? * If you're having trouble defining your solutions store, then think about it in terms of the base case . - The answer may be larger than a -bit integer.

Sample Input 0

```
4 3
1 2 3
```

Sample Output 0

```
4
```

Explanation 0

There are four ways to make change for using coins with values given by :

- 1.
- 2.
- 3.
- 4.

Sample Input 1

```
10 4
2 5 3 6
```

Sample Output 1

```
5
```

Explanation 1

There are five ways to make change for units using coins with values given by :

- 1.
- 2.
- 3.
- 4.
- 5.

[Change Theme](#)

Language

Python 3



```
1  import sys
2
3  def getWays(n, c):
4      dp = [0] * (n + 1)
5      dp[0] = 1
6
7      for coin in c:
8          for i in range(coin, n + 1):
9              dp[i] += dp[i - coin]
10
11     return dp[n]
12
13 if __name__ == '__main__':
14     input_data = sys.stdin.read().split()
15     if input_data:
16         n = int(input_data[0])
17         m = int(input_data[1])
18         c = list(map(int, input_data[2:]))
19
20     res = getWays(n, c)
21     print(res)
22
```

Line: 22 Col: 1



Upload Code as File



Test against custom input

Run Code

Submit Code

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

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✓ Sample Test case 1

```
1 | 4 3
2 | 1 2 3
```

Your Output (stdout)

```
1 | 4
```

Expected Output

```
1 | 4
```

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